The Divide (Mountain Pass / Strategic Chokepoint Warfare)

Executive Summary

The Divide is a **high-altitude mountain-pass battlefield** built for **strategic chokepoint warfare, elevation-driven combat, autonomous route control, and multi-domain coordination across narrow, high-risk terrain**. The environment features sheer cliffs, winding passes, avalanche-scarred slopes, fortified ridgelines, collapsed tunnels, old military outposts, and steep switchback roads. The map emphasizes Obsidian Protocol's autonomous elevation-aware routing, sensor-fusion under mountain fog and rock occlusion, comms-relay warfare, and logistics under severe terrain constraints. In AR (Nreal Light/Air), the map anchors at ~4–5 m viewing distance, covering a 10x10 m physical area representing ~1 km^2 of mountain-pass terrain. Key features include chokepoint control, overwatch ridges, ambush corners, and high-risk mobility corridors.

Design Goals & Gameplay

Chokepoint-Focused Warfare

The Divide prioritizes control of narrow passes, switchbacks, and cliffside trails. Holding key chokepoints determines reinforcement flow and strategic dominance.

Elevation-Driven Combat

High-ground ridges provide overwatch for snipers, artillery, and drones. Verticality shapes sightlines, movement, and engagement ranges.

Ambush & Route Denial

Blind corners, collapsed tunnels, and narrow ledges create ideal ambush zones. Route denial becomes a core strategic tool.

Autonomy & Terrain-Aware Routing

Autonomous units choose routes based on elevation advantage, rockfall hazards, fog density, and sensor clarity. Units adapt to blocked roads, unstable slopes, and shifting mountain conditions.

Sensor & Information Warfare

Fog, dust, rock occlusion, thermal plumes, and elevation clutter degrade sensors. Relay towers on ridges restore clarity. Destroying relays creates blind pockets across the pass.

Logistics & Resources

Fuel caches, mountain cabins, and old military outposts serve as resupply nodes. Convoys must navigate narrow roads and avoid ambush-prone switchbacks.

Deployment & Progress

A fixed deployment budget (~10,000 points) forces players to choose between fast armor, overwatch specialists, or stealth mountain infantry. Blue deploys from the southwest lower-pass gate; Red deploys from the northeast ridge-top fortress.

Spatial Constraints (Nreal AR)

Device Capabilities

Nreal Light/Air: ~53° diagonal FOV, 1920×1080 per-eye resolution. Optimal viewing distance ~4–5 m.

Playable Area & Scaling

10×10 m physical area ≈ 1 km^2 mountain-pass zone (1:100 scale). Designed for tabletop AR.

Tracking & Occlusion

Inside-out tracking detects horizontal planes. Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view; assume minimal real-world occlusion.

Layout Overview

Coordinate System

1000×1000 grid (1 unit ≈ 1 m). Origin (0,0) = southwest corner.

Key Locations

Blue HQ (Lower-Pass Gate Command) — (0,0), 150x150 m Armored vehicles, drone pads, mountain staging.
Red HQ (Ridge-Top Fortress) — (1000,1000), 150x150 m Elevated bunker with radar mast, AA emplacements, and overwatch.
Central Pass Corridor — (500,450), 300x300 m Narrow pass, steep cliffs, fog pockets, long-range overwatch.
Abandoned Outpost Cluster — (250,800), 120x120 m Ruins, generators, interior flanking routes.
Ridge Network — (800,300), 200x200 m High-ground overwatch, sniper nests, drone launch pads.
Switchback Chokepoint — (0,450 → 400,900) Primary chokepoint; narrow, exposed, ideal for ambushes.
Mountain Trail Route — (300,0 → 450,350) Safer but slow; rock hazards and limited cover.
Ridge Relay A — (650,150), +240 m Long-range sensor vantage point.
Ridge Relay B — (150,650), +260 m Comms relay hub and overwatch control node.
Neutral Forward Outpost — (500,950) Drone resupply zone.

Chokepoints & Sightlines

Switchback Chokepoint

Major chokepoint; controlling it determines convoy movement.

Ridge Network

Extreme long-range sightlines; ideal for overwatch and artillery.

Pass Corridor

Long-range engagements; fog and rock occlusion affect accuracy.

Mountain Trail Route

Predictable but safe; vulnerable to ridge overwatch.

Sightlines

Ridge Relay A provides long-range sensor coverage (~350 m). Fog and dust create fog-of-war pockets.

Deployment Zones

Blue Deployment: SW lower-pass gate + armor staging.
Red Deployment: NE ridge-top fortress + AA emplacements.
Neutral Forward Zone: Outpost at (500,950) for drone drops.

Assets & Environment

Terrain

Cliffs, ridges, passes, switchbacks, outposts, mountain trails.

Vegetation

Sparse shrubs, pine trees, alpine grass.

Buildings & Props

Military: AA guns, radar dishes, bunkers. Rural: Cabins, fences, wooden structures. Logistics: Trucks, crates, fuel drums, drones.

Effects

Fog, dust, wind gusts, rockfall debris.

Audio

Wind, distant machinery, rock impacts, radio chatter.

LOD & Budget

Asset Cat	Count	Tri/Item	Total Tris
Mountain	3	200k	600k
Industrial	4	80k	320k
Buildings	15	50k	750k
Roads/Pa	4	30k	120k
Props	50	10k	500k
Total	—	—	**~2.29M**

Texture Budget

2048^2 for terrain/buildings, 1024^2 for props. DXT1/5 compression. Target ≤100 MB.

Unreal / Nreal Integration

Engine & Plugins

Unreal ARTemplate + Nreal SDK (XREAL). ARKit/ARCore enabled.

Rendering

Forward renderer, stationary lights, baked GI, aggressive culling.

Anchors

Single spatial anchor for entire map.

Interaction

Phone pointer for pan/zoom; HUD as screen-space widgets.

Multiplayer

Client-server architecture; offload AI/physics to server.

Implementation Plan & Testing

Milestones

1. **Prototype Layout — Month 1** Block-out pass corridor, ridges, outposts, AR anchor test.
2. **Core Systems — Month 2** NavMesh, chokepoint autonomy behaviors, elevation logic.
3. **AR Integration — Month 3** Anchors, plane detection, basic interactivity.
4. **Content Fill — Month 4–5** Final models, textures, lighting, audio.
5. **Gameplay & Polish — Month 6–7** Objectives, balance, VFX (fog, dust, rockfall).
6. **Testing & QA — Month 8** Multiplayer stress test, sensor validation.

Testing Plan

Autonomy

Verify chokepoint navigation, elevation-aware pathfinding, hazard avoidance.

Comms/Sensors

Test relay destruction, fog-of-war updates, LOS checks under mountain occlusion.

Performance

≥60 Hz AR render on target device.

QA Checklist

Navigation, chokepoint logic, deployment budget enforcement, anchor drift, sensor accuracy.